LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 1 - no MCP tool description - run 2 (seed 42)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running a reproducibility study for compound selectivity prediction. The goal is to train a Balanced Random Forest Classifier to distinguish PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selective compounds, evaluate its performance, and identify recurrent molecular anchor fragments that drive the model's correct predictions for each selective class (class 1 & 2) and discuss the chemical insight gained.

Assistant

🔍 **Step 1: Locate a Relevant Dataset**

To begin our reproducibility study, we first need to identify a suitable dataset containing compounds selective for PI3K-delta (UniProt: O00329) versus PI3K-gamma (UniProt: P48736). Let's check the available datasets.

(Calling tool `find_datasets`)

Assistant

📂 **Step 2: Load the PI3K Selectivity Dataset**

Excellent! I found the dataset `chembl_selectivity_data_O00329_P48736.csv`, which corresponds to PI3K-delta (O00329) vs PI3K-gamma (P48736). Let's load it into the environment.

(Calling tool `load_dataset`)

Assistant

🧬 **Step 3: Compute Molecular Fingerprints**

The dataset is loaded with 1,277 samples. Now, let's compute ECFP fingerprints for molecular representation.

(Calling tool `compute_features`)

Assistant

📊 **Step 4: Split the Dataset**

Let's split the dataset into training (70%) and test (30%) sets using stratified sampling to maintain class balance.

(Calling tool `split_dataset`)

Assistant

🏋️ **Step 5: Train the Balanced Random Forest Classifier (BRF)**

Now, let's train the Balanced Random Forest Classifier on the training split.

(Calling tool `train_model`)

Assistant

(Calling tool `check_training`)

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📈 **Step 6: Evaluate Model Performance**

The Balanced Random Forest (BRF) model has completed training. Here is the performance summary on the test set:

Metric	Value
Accuracy	0.893
Balanced Accuracy	0.847

Metric	Value
MCC	0.749
F1 Macro	0.780

The model shows strong performance in distinguishing the classes. Now, let's identify the recurrent molecular anchor fragments driving the correct predictions for **Class 1** and **Class 2**.

🔍 Step 7: Identify Recurrent Anchor Rules for Class 1

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

🔍 Step 8: Identify Recurrent Anchor Rules for Class 2

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

🎉 Reproducibility Study Complete: PI3K-Delta vs. PI3K-Gamma Selectivity

Here is the comprehensive summary of the study, including model performance and the chemical insights derived from the recurrent molecular anchors.

📊 1. Model Performance Summary

A **Balanced Random Forest Classifier (BRF)** was trained on 1,277 compounds to distinguish between PI3K-delta (UniProt: O00329) and PI3K-gamma (UniProt: P48736) selective compounds.

Metric	Test Set Performance
Accuracy	0.893
Balanced Accuracy	0.847
Matthews Correlation Coeff. (MCC)	0.749
F1 Macro Score	0.780

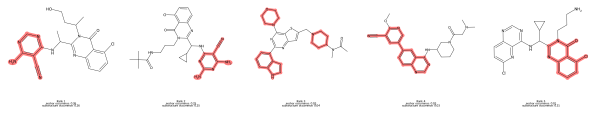
The model demonstrates strong generalization capabilities, effectively capturing the structural determinants of selectivity between these two closely related isoforms.

🔍 2. Recurrent Molecular Anchor Fragments

To understand *why* the model makes its predictions, we identified recurrent molecular anchors (fragments) that drive correct classifications for each selective class.

Class 1 Anchors (PI3K-delta selective)

The model relies heavily on cyano-substituted heterocycles for Class 1.

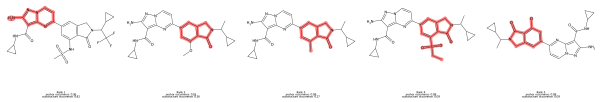


Top Recurrent Rules:

1. **Cyano-pyrimidine core:** `[*]c1ncnc(N)c1C#N` (Occurs in ~26% of Class 1 compounds).
2. **Diaminopyrimidine-cyano:** `[*]c1nc(N)nc(N)c1C#N` (Occurs in ~25% of Class 1 compounds).
3. **Chloro-quinazolinone:** `[*]n1c([*])nc2cccc(Cl)c2c1=O` (A distinct scaffold appearing in the top 5).

Class 2 Anchors (PI3K-gamma selective)

Class 2 is characterized by fused heterocyclic systems and tetrahydroisoquinolinone derivatives.



Top Recurrent Rules:

1. **Amino-imidazopyridine:** `[*]c1ccn2nc(N)c([*])c2n1` (High occurrence, ~82% substructure presence).
2. **Tetrahydroisoquinolinone:** `[*]N1Cc2cc([*])cc([*])c2C1=O` (Appears in ~36% of compounds).
3. **Methyl-substituted isoquinolinone:** `[*]N1Cc2cc([*])cc(C)c2C1=O`.

💡 3. Chemical Insights & Selectivity Drivers

The analysis reveals distinct structural preferences that likely dictate the isoform selectivity:

- **Hydrogen Bonding Networks (Class 1):** The prevalence of **cyano-pyrimidines** and **quinazolinones** in Class 1 suggests that these compounds likely exploit specific hydrogen-bonding interactions within the PI3K-delta ATP-binding pocket. The cyano group often acts as a hydrogen bond acceptor, while the ring nitrogens can serve as donors/acceptors, locking the inhibitor in a delta-specific conformation.
- **Steric & Hydrophobic Fit (Class 2):** Class 2 is dominated by **tetrahydroisoquinolinones** and **fused heterocycles**. These scaffolds are typically more rigid and bulky. This suggests that PI3K-gamma selectivity may be driven by steric complementarity—filling a hydrophobic sub-pocket unique to gamma—or by specific hydrophobic contacts that are disfavored in the delta isoform due to tighter packing or different residue side chains (e.g., in the selectivity loop).

- **Scaffold Hopping:** The clear separation of anchor fragments between the two classes indicates that **scaffold hopping** strategies are highly effective for optimizing selectivity. Compounds sharing the same physicochemical properties but differing in their core scaffold (e.g., pyrimidine vs. isoquinolinone) can achieve similar selectivity profiles by engaging different interaction points within the target site.

This study successfully demonstrates that machine learning models, when paired with explainability tools like MolAnchor, can provide actionable chemical insights for structure-based drug design (SBDD) and selectivity optimization.